IBM QRADAR · AQL CHEAT SHEET

Firewall Log Analysis & Threat Hunting

Nine ready-to-use AQL queries built on native QRadar fields and functions.

تسعة استعلامات AQL جاهزة لتحليل سجلات الجدار الناري وصيد التهديدات.

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Before you start: replace `'FW-01'` / `'VPN-01'` with your own log source names. List them with: `SELECT LOGSOURCENAME(logsourceid) AS log_source, COUNT(*) FROM events GROUP BY logsourceid LAST 24 HOURS`

01 Top Blocked Source IPs

Most frequently denied source IPs — who is knocking on your perimeter.

تحديد عناوين IP الأكثر تعرضاً للحظر من الجدار الناري.

```
SELECT sourceip, COUNT(*) AS attempts
FROM events
WHERE LOGSOURCENAME(logsourceid) = 'FW-01'
  AND CATEGORYNAME(category) = 'Firewall Deny'
GROUP BY sourceip
ORDER BY attempts DESC
LAST 7 DAYS
```

02 Unusual Destination Ports (Outbound)

Allowed outbound connections to non-standard ports — possible C2, tunneling, or shadow IT.

اتصالات صادرة مسموحة إلى منافذ غير معتادة، قد تدل على C2 أو أنفاق.

```
SELECT sourceip, destinationip, destinationport, COUNT(*) AS connections
FROM events
WHERE LOGSOURCENAME(logsourceid) = 'FW-01'
  AND CATEGORYNAME(category) = 'Firewall Permit'
  AND eventdirection = 'L2R'
  AND destinationport NOT IN (80, 443, 22, 53, 25, 21)
GROUP BY sourceip, destinationip, destinationport
ORDER BY connections DESC
LAST 7 DAYS
```

03 Port Scanning Detection

Sources hitting a high number of distinct destination ports.

كشف محاولات فحص المنافذ (Port Scanning) من مصدر واحد.

```
SELECT sourceip, UNIQUECOUNT(destinationport) AS unique_ports,
       COUNT(*) AS attempts
FROM events
WHERE LOGSOURCENAME(logsourceid) = 'FW-01'
GROUP BY sourceip
HAVING UNIQUECOUNT(destinationport) > 15
ORDER BY unique_ports DESC
LAST 24 HOURS
```

04 Outbound Connection Spikes (Hourly)

Internal hosts generating an abnormal volume of outbound connections per hour.

رصد الأجهزة الداخلية ذات الارتفاع غير الطبيعي في الاتصالات الصادرة كل ساعة.

```
SELECT sourceip,
       DATEFORMAT(starttime, 'yyyy-MM-dd HH:00') AS hour_bucket,
       COUNT(*) AS connections
FROM events
WHERE LOGSOURCENAME(logsourceid) = 'FW-01'
  AND eventdirection = 'L2R'
GROUP BY sourceip, hour_bucket
HAVING COUNT(*) > 1000
ORDER BY connections DESC
LAST 7 DAYS
```

Note: eventdirection values: L2R (local → remote), R2L (remote → local), L2L, R2R.

05 High-Severity Events

Events QRadar rates as high severity. severity is numeric (0–10), not text.

عرض الأحداث عالية الخطورة. حقل severity رقمي من 0 إلى 10.

```
SELECT sourceip, CATEGORYNAME(category) AS category_name,
       MAX(severity) AS max_severity, COUNT(*) AS events
FROM events
WHERE LOGSOURCENAME(logsourceid) = 'FW-01'
  AND severity >= 7
GROUP BY sourceip, category
ORDER BY events DESC
LAST 7 DAYS
```

06 After-Hours VPN Logins

Successful VPN logins outside business hours (22:00–05:59).

كشف اتصالات VPN الناجحة خارج ساعات الدوام.

```
SELECT username, sourceip,
       DATEFORMAT(starttime, 'yyyy-MM-dd HH:mm') AS login_time
FROM events
WHERE LOGSOURCENAME(logsourceid) = 'VPN-01'
  AND CATEGORYNAME(category) ILIKE '%login succe%'
  AND DATEFORMAT(starttime, 'HH') IN
      ('22','23','00','01','02','03','04','05')
ORDER BY starttime DESC
LAST 7 DAYS
```

Note: DATEFORMAT follows the QRadar console time zone — adjust the hour list if needed.

07 Inbound Allowed Traffic from Unexpected Countries

Allowed inbound connections from outside your expected geography.

اتصالات واردة مسموحة من دول غير معتادة.

```
SELECT sourceip, GEO::LOOKUP(sourceip, 'country') AS country,
       destinationport, COUNT(*) AS connections
FROM events
WHERE LOGSOURCENAME(logsourceid) = 'FW-01'
  AND CATEGORYNAME(category) = 'Firewall Permit'
  AND eventdirection = 'R2L'
  AND GEO::LOOKUP(sourceip, 'country') <> 'Saudi Arabia'
GROUP BY sourceip, destinationport
ORDER BY connections DESC
LAST 7 DAYS
```

08 Authentication Failure Patterns

Repeated login failures per user and source — brute force or credential stuffing.

تحليل فشل محاولات الدخول المتكررة لكل مستخدم ومصدر.

```
SELECT sourceip, username, COUNT(*) AS failures
FROM events
WHERE LOGSOURCENAME(logsourceid) IN ('FW-01', 'VPN-01')
  AND CATEGORYNAME(category) ILIKE '%login fail%'
GROUP BY sourceip, username
HAVING COUNT(*) > 5
ORDER BY failures DESC
LAST 24 HOURS
```

09 Traffic to Known Malicious IPs

Connections to IPs in a threat-intel reference set (X-Force, MISP, or your own IoC feed).

ربط حركة الشبكة بعناوين خبيثة معروفة من مصادر Threat Intelligence.

```
SELECT sourceip, destinationip, destinationport, COUNT(*) AS hits
FROM events
WHERE LOGSOURCENAME(logsourceid) = 'FW-01'
  AND REFERENCESETCONTAINS('Malicious IPs', destinationip)
GROUP BY sourceip, destinationip, destinationport
ORDER BY hits DESC
LAST 7 DAYS
```

Note: Replace 'Malicious IPs' with the name of your reference set.

Implementation Notes

- **Category names** (Firewall Deny, Firewall Permit, login success/failure) are QRadar normalized low-level categories, so they work across firewall vendors.
- **Thresholds** (> 15, > 1000, > 5) are starting points — baseline your environment and tune them to reduce false positives.
- **Performance:** always filter by log source and keep time windows tight on busy environments.

github.com/omar-alghamdi-tech/qradar-aql-firewall-cheatsheet